

POWER GENERATION ON HIGHWAY BY USING VERTICAL AXIS

WIND TURBINE

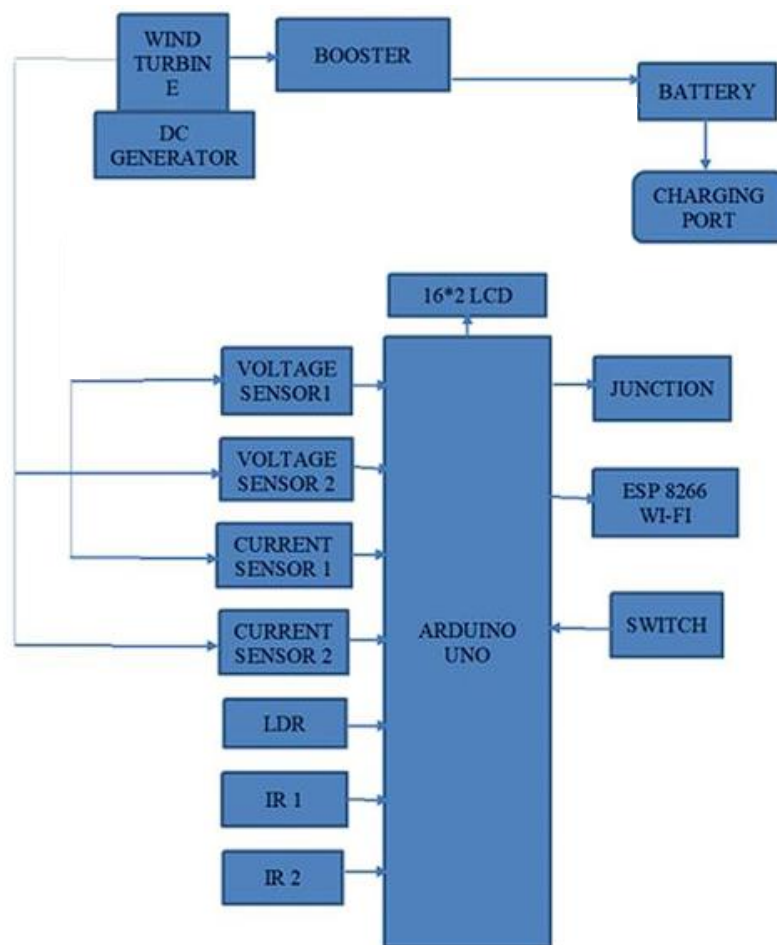


Fig 1 : Block Diagram

ABSTRACT -

This project focuses on use of air (wind) on Highway divider with the help of VAWT. When the vehicle passes on highway it produces a considerable amount of air (wind) due to its speed. This wind strikes, tangentially, on the blade of wind turbine and makes rotation of turbine in one-direction. The electrical output of turbine is stored in Battery, which can be further used for Street lighting (related usage).

INTRODUCTION

The main motive behind this project is to design a Vertical axis Wind Turbine which effectively uses the wind Energy, generated by Vehicle speed on highway.

OBJECTIVE

- A) The main aim of this project is to reduce pollute produced while burning fossil fuel.
- B) To utilize the unused pressurized air to wind turbine, from which the kinetic Energy of turbine is converted to Electrical Energy.
- C) To reduce the Electrical pressure on the GRIDS.

EQUIPMENTS REQUIRED

1. Wind Turbine + Gear + DC Generator
2. Charge Controller
 - Boost converter
 - Arduino UNO
 - LCD
 - Dump load
 - Voltage and Current sensor
3. Battery- {12V , 5200mAh}

WORKING PRINCIPLE

The moving vehicle on highway may be all types such as small or heavy vehicles. Whenever vehicle moves on both side of the highway divider then some pressurized air is produced due to the speed of vehicle. This pressurized air is strike on the blade of vertical axis wind turbine and turbine makes a rotation. The shaft of the vertical axis wind turbine is Connected to generator with the help of gear mechanism. The generated electricity is stored in the battery. The stored energy used as a street lighting.

CONCLUSION

Vertical Axis Wind Turbines (VAWTs) are a low-cost, environmentally-friendly option that can be used for small-scale operations and maintenance. The use of vertical windmills on highways can provide an affordable alternative to traditional power sources, especially in remote areas where the cost of grid connectivity is high. By installing vertical windmills on highways, we can create a network of microgrids that can supply power to nearby communities and industries.

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